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LUMIN

“VISION ASSISTANCE FOR VISUALLY IMPAIRED”

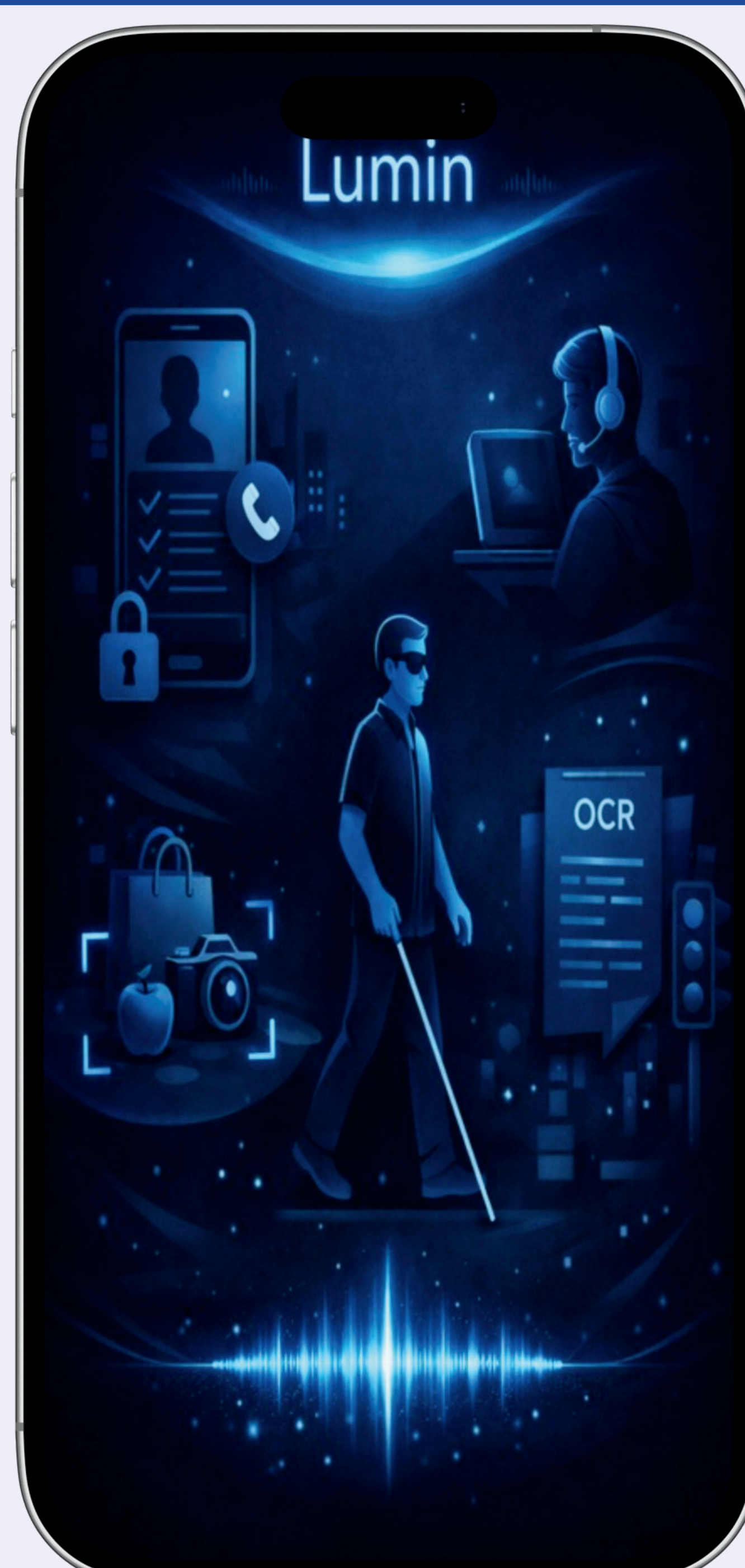


1 | PROBLEM

Visually impaired individuals struggle to navigate indoor spaces safely. Existing tools either require human help, special hardware, or only solve one piece of the puzzle (like reading text or naming objects) — leaving users without continuous awareness of their surroundings.

3 | SOLUTION

LUMIN is mobile app that uses a smartphone's camera and microphone to perceive the environment and deliver clear audio feedback. It detects nearby objects, estimates proximity, reads printed text on demand, and responds to natural voice commands — helping users move through indoor spaces more safely and independently.



2 | OBJECTIVE

To design a smartphone-based assistive system that gives blind and low-vision users real-time indoor awareness through voice interaction, object and distance estimation, and text recognition — all without requiring specialized hardware.

4 | USED TECHNOLOGY

LUMIN combines a custom Roboflow-collected dataset of Egyptian indoor scenes, used to fine-tune YOLO for region-specific object classes, with Google Cloud Vision API for OCR, OpenAI's Realtime Voice Agent for natural-language reasoning and conversation, and Picovoice for hands-free wake-word activation — all coordinated through a cloud backend for fast, reliable performance.



5 | STAKEHOLDERS

The primary users are blind and low-vision individuals seeking safer, more independent indoor mobility.

6 | REAL-WORLD USE CASES

LUMIN helps users detect nearby objects, estimating how close they are. It also reads printed and on-screen text aloud — including medicine labels, product packaging, books, and screens — giving users quick, voice-guided access to information they'd otherwise need sight for.

